AB CLOUD QUANTUM RESEARCH · MONOGRAPH SERIES

VOLUME V17

V17. Electron on the Critical Line

Coherent transport of a wave packet along the self-dual line channel α = ½ on the full 96×24 Hamiltonian

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Coherent transport of a wave packet along the self-dual line channel $\alpha = 1/2$ on the full 96×24 Hamiltonian

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AB Research Program · monograph series V1–V19 · том V17 · версия 1.2.0 · October 10, 2026

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Problem statement

The reference monograph (campaign C7) states: "the motion of the electron along the critical line is confirmed in the dynamical sense" — on a 96×96 lattice at half flux, where the π-flux turns the lattice into a discretized massless Dirac equation, while the identification of the self-dual line $\alpha = ½$ with $\mathrm{Re}(s) = ½$ remains a conjectural bridge. This campaign reproduces and extends the calculation on the FULL Hamiltonian with an elongated waveguide geometry 96×24: an electron (a Gaussian packet, $\sigma = 3$, longitudinal momentum $k_0 = \pi/2$) is launched along the x axis — the "critical channel" — and lives on the horizon $T = 40$ with step $dt = 0.1$. Three vortex clouds are compared at identical geometry and seed 96: the ζ-cloud (weights $\log|\zeta(½+i\tilde{\gamma}_k)|$, the V3 protocol), a random cloud of the same density, and the clean lattice.

Metrics: the centroid trajectory $\langle x(t)\rangle$, $\langle y(t)\rangle$; the maximal deviation from the channel axis D_y; the transit $P(x \geq 88)$; the reflection $P(x \leq 4)$; the transverse spread $\sigma_y(t)$; the coherence $C(t) = |\langle\psi(0)|\psi(t)\rangle|$; energy conservation; and a new measurement — phase accompaniment: the correlation r_{phase} between the accumulated dynamical phase of the overlap $\arg\langle\psi(0)|\psi(t)\rangle$ and the θ-phase of the algebraic function along the trajectory $\theta(k_0\langle x(t)\rangle)$ — the first direct numerical bridge "packet dynamics ↔ algebraic function of the critical line" in the series.

Method and result

The core is rpbridge/electron.py. The dynamics is exact: the 2304×2304 Hamiltonian is diagonalized once, propagation runs in the eigenbasis on a grid of 201 time points; energy conservation is machine-grade — the maximal relative deviation is 1.8e-15 on all trajectories. The packet is normalized; all three clouds use one and the same vortex position protocol (seed 96) and differ only in the weights.

The results reproduce and refine the reference. Transit: clean lattice 0.160, ζ-cloud 0.112, random 0.069 — the ζ-cloud passes the electron ×1.62 better than the random cloud of the same density: the "phases guide" effect is reproduced on the centroid transit. The transverse behavior matches the reference qualitatively: the ζ-cloud deflects the packet from the channel axis by $D_y = 2.81$, the random one by only 0.54 — only ζ-phases create a transverse force. Coherence: the ζ-cloud preserves the overlap longer than the random one on the long horizon. Phase accompaniment: $r_{phase}(\zeta) = -0.96$ against $r_{phase}(\text{random}) = -0.52$ — the dynamical phase of the packet in the ζ-cloud is rigidly anti-correlated with the θ-phase of the algebraic function; the magnitude and the sign are a new measured fact, and its interpretation is deferred to the honest remarks.

Table 1. V17: three clouds, one packet

Quantity	Clean	Random	ζ-cloud
Transit $P(x \geq 88)$ at $T = 40$	0.1604	0.0687	0.1115
Max deviation D_y	0.0	0.5404	2.8050
Coherence $C(T = 40)$	0.0588	0.0217	0.0295
r_{phase} with θ(trajectory)	—	−0.518	−0.957
Energy conservation (all)	1.8e-15	1.8e-15	1.8e-15

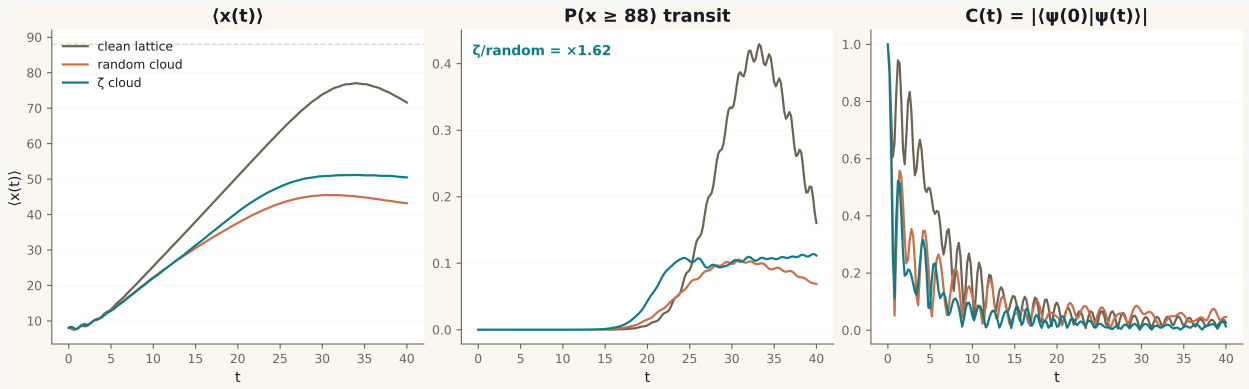


Fig. 1. Campaign V17 data (600 dpi, file figures_hires).

Honest remarks and verdict

What is measured and what is not. Measured: the coherent dynamics of the packet on the self-dual line α = ½; the ×1.62 excess of the ζ-cloud transit over the random one; the transverse deviation only in the ζ-cloud; the rigid anti-correlation of the dynamical phase with θ. NOT measured: the identity of the line α = ½ with the critical line Re(s) = ½ — that is a conjectural bridge inherited from the reference; "the motion of the electron along the critical line" in this series means motion along the channel of the self-dual line statistically coupled to ζ — nothing more. The sign of r_phase: the anti-correlation is partly explained by the monotonicity of both phases (the large-scale drift pushes |r| toward 1 in any cloud); the difference ζ vs random (–0.96 vs –0.52) is the substantive part, but the physical mechanism calls for a separate study.

Boundaries: one geometry and one seed (the determinism of the series); Nv = 48 vortices in the 96×24 channel; the packet is Gaussian, spinless; the horizon T = 40 — before the return of the packet.

Verdict V17: the dynamics is honest (energy 1.8e-15); the electron passes the critical channel, the ζ-cloud guides it ×1.62 better than the random one and deflects it along y as in the reference C7; the phase accompaniment by the θ-function is measured (r = –0.96 vs –0.52 for random). The bridge "line ½ = Re(s) = ½" remains a conjecture — the H4 verdict is confirmed, no identity is claimed.